Commutative Algebra

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Contents

Intro

These are my own personal notes in commutative algebra, a course I will hopefully be taking in the fall of 2026. Note that as these are for myself there might be gaps, inconsistencies and whatnot which might make it hard to follow so keep that in mind if you are considering using my notes for yourself.

1 Rings

Throughout the course, and these notes, a ring means a commutative ring with unity (meaning multiplicative identity). So every ring R has distinguished elements $1_R, 0_R \in R$ and multiplication satisfies $ab = ba$.

Definition 1.1 ((Commutative Unital) Ring). *A ring is a set A together with binary operations $+, \cdot : A \times A \to A$ such that*

1. *$(A, +)$ is an abelian group with identity 0,*
2. *multiplication is associative,*
3. *multiplication is commutative,*
4. *multiplication distributes over addition,*
5. *there exists $1 \in A$ such that $1a = a$ for all $a \in A$.*

Example 1.1. *Examples of familiar rings include*

$$\mathbb{Z}, \quad \mathbb{Q}, \quad \mathbb{R}, \quad \mathbb{C}$$

with the usual addition and multiplication.

If A is a ring, then the polynomial ring $A[x]$ is also a ring. More generally, $A[x_1, \dots, x_n]$ is the ring of polynomials in n commuting variables with coefficients in A.

Definition 1.2 (Ring homomorphism). *Let A and B be rings. A function*

$$\phi : A \to B$$

is a ring homomorphism if

1. *$\phi(1_A) = 1_B$,*
2. *$\phi(a + a') = \phi(a) + \phi(a')$,*
3. *$\phi(aa') = \phi(a)\phi(a')$.*

If ϕ is a bijection, then it is called an isomorphism.

The condition $\phi(1_A) = 1_B$ is a convention that one may or may not choose, but in this course it is not optional, and we say all homomorphisms must explicitly satisfy this.

Proposition 1.1. *Let A be a ring. There exists a unique ring homomorphism*

$$\iota : \mathbb{Z} \to \mathbb{A}.$$

It is given by

$$\iota(n) = n \cdot 1_A.$$

Proof. Since ι is a rinh homomorphism, it must satisfy

$$\iota(1) = 1_A.$$

Then additivity forces

$$\iota(2) = \iota(1+1) = \iota(1) + \iota(1) = 1_A + 1_A,$$

and similarly

$$\iota(n) = \sum_{k=1}^{n} 1_A = \underbrace{1_A + \cdots + 1_A}_{\text{n times}}$$

for $n > 0$. Also,

$$\iota(-n) = -\iota(n).$$

This there is at most one such homomorphism. It is straightforward to check that the function $n \mapsto n \cdot 1_A$ preserves addition, multiplication, and sends 1 to 1_A. Hence it is a ring homomorphism. □

Definition 1.3 (Characteristic). *The characteristic of a ring A, denoted* char(A), *is the unique integer $n \geq 0$ such that*

$$\operatorname{ker}(\mathbb{Z} \to A) = (n).$$

Example 1.2. *Here are the characteristics of a few familiar rings:*

- char$(\mathbb{Z}) = 0$,
- char$(\mathbb{F}_p) = p$,
- char$(\mathbb{Q}) = 0$.

Intuitively, the characteristic is the smallest $n \in \mathbb{Z}^+$ such that $n \cdot 1_A = 0_A$ with $n := 0$ if no such n exists, and in fact, the characteristic is usually defined as such in a first course in abstract algebra.

Proposition 1.2 (Universal property of $\mathbb{Z}$-polynomials). *Let A be a ring and let $a \in A$. There exists a unique ring homomorphism*

$$\phi_a : \mathbb{Z}[x] \to A$$

such that

$$\phi_a(x) = a.$$

It is given by

$$\phi_a\left(\sum_{k=0}^{n} c_k x^k\right) = c_0 \cdot 1_A + c_1 a + \cdots + c_n a^n.$$

Proof. A ring homomorphism $\phi : \mathbb{Z}[x] \to A$ must agree with the unique homomorphism $\mathbb{Z} \to A$ on constant polynomials.

If $\phi(x) = a$, then multiplicativity forces

$$\phi(x^n) = a^n.$$

Additivity then forces

$$\phi(c_0 + c_1 x + \cdots + c_n x^n) = c_0 \cdot 1_A + c_1 a + \cdots + c_n a^n.$$

Thus there is at most one such homomorphism.

Conversely, the formula above defines a ring homomorphism, by the usual rules for polynomial addition and multiplication. □

So homomorphism $\mathbb{Z}[x] \to A$ are the same thing as choices of an element $a \in A$. Intuitively, $\mathbb{Z}[x]$ is the "freest" ring generated by one element.

Definition 1.4 (Ideal). *Let A be a ring. An ideal $I \subseteq A$ is a subset such that:*

1. *I is an additive subgroup of A,*
2. *for every $a \in A$ and every $x \in I$, we have*

$$ax \in I.$$

Definition 1.5 (Principal ideal). *Let A be a ring and $a \in A$. The principal ideal by a is*

$$(a) = \{ra \mid r \in A\}.$$

An ideal is called principal if it is of the form (a) for some $a \in A$.

More generally, if $a_1, \ldots, a_n \in A$, then

$$(a_1, \ldots, a_n) = \{r_1 a_1 + \cdots + r_n a_n \mid r_i \in A\}.$$

This is the smallest ideal containing all of $a_1, \dots, a_n$.

Example 1.3. *$(x, y) \subseteq k[x, y]$ is such an ideal.*

$$(x, y) = \{f(x, y)x + g(x, y)y \mid f, g \in k[x, y]\}.$$

It is not the whole ring. It consists precisely of polynomials with zero constant term:

$$(x, y) = \{h \in k[x, y] \mid h(0, 0) = 0\}.$$

Example 1.4. *$(0) = \{0\}$ is a (very uninteresting) ideal. $(1) = A$ is another (very uninteresting) ideal. For $n \in \mathbb{Z}$ we have that $(n) = n\mathbb{Z}$ is an ideal of $\mathbb{Z}$.*

Proposition 1.3. *Let $\phi : A \to B$ be a ring homomorphism. Then*

$$\ker \phi = \{a \in A \mid \phi(a) = 0\}$$

is an ideal of A.

Proof. First we show that $\ker \phi$ is an additive subgroup. Since $\phi(0) = 0$, we have $0 \in \ker \phi$. If $x, y \in \ker \phi$, then

$$\phi(x - y) = \phi(x) - \phi(y) = 0 - 0 = 0.$$

Hence $x - y \in \ker \phi$. Now let $a \in A$ and $x \in \ker \phi$. Then

$$\phi(ax) = \phi(a)\phi(x) = \phi(a) \cdot 0 = 0.$$

Thus $ax = \ker \phi$. Therefore $\ker \phi$ is an ideal. □

Definition 1.6 (Quotient ring). *Let $I \subseteq A$ be an ideal. The quotient ring A/I is the set of additive cosets*

$$a + I$$

with addition and multiplication defined by

$$(a + I) + (b + I) = (a + b) + I$$

and

$$(a + I)(b + I) = ab + I.$$

Proposition 1.4. *If $I \subseteq A$ is an ideal, then A/I is a ring.*

Proof. The quotient A/I is already an abelian group under addition because I is an additive subgroup.

We need to check multiplication is well-defined.

Suppose

$$a + I = a' + I$$

and

$$b + I = b' + I.$$

Then

$$a - a' \in I, \qquad b - b' \in I.$$

We want to prove

$$ab + I = a'b' + I,$$

i.e.

$$ab - a'b' \in I.$$

Compute:

$$ab - a'b' = ab - a'b + a'b - a'b' = (a - a')b + a'(b - b').$$

Since $a - a' \in I$, and I absorbs multiplication by elements of A, we have

$$(a - a')b \in I.$$

Similarly,

$$a'(b - b') \in I.$$

Since I is closed under addition,

$$ab - a'b' \in I.$$

Therefore multiplication is well-defined.

The ring axioms for A/I then follow from the ring axioms in A. $\square$

The quotient map is

$$\pi : A \to A/I, \;\; a \mapsto a + I.$$

Its kernel is exactly I:

$$\ker \pi = I.$$

Theorem 1.1 (Fundamental homomorphism theorem). *Let*

$$\phi : A \to B$$

be a ring homomorphism. Then

$$\operatorname{im} \phi$$

is a subring of B, and there is an isomorphism

$$A/\ker \phi \cong \operatorname{im} \phi$$

given by

$$a + \ker \phi \mapsto \phi(a).$$

Proof. Define

$$\overline{\phi} : A/\ker \phi \to \operatorname{im} \phi$$

by

$$\overline{\phi}(a + \ker \phi) = \phi(a).$$

We first check well-definedness.

Suppose

$$a + \ker \varphi = b + \ker \varphi.$$

Then

$$a - b \in \ker \varphi.$$

Thus

$$\varphi(a - b) = 0.$$

So

$$\varphi(a) - \varphi(b) = 0,$$

and hence

$$\varphi(a) = \varphi(b).$$

Therefore $\overline{\varphi}$ is well-defined.

It is a ring homomorphism because φ is.
It is surjective onto $\operatorname{im}\varphi$ by definition.
Finally, its kernel is trivial. Indeed,

$$\overline{\varphi}(a + \ker\varphi) = 0$$

if and only if

$$\varphi(a) = 0,$$

if and only if

$$a \in \ker\varphi,$$

if and only if

$$a + \ker\varphi = 0 + \ker\varphi.$$

Thus $\overline{\varphi}$ is injective. Hence it is an isomorphism. $\square$

Example 1.5.

$$\mathbb{Z}[x]/(x^2+1) \cong \mathbb{Z}(i) = \{a + bi \mid a, b \in \mathbb{Z}\}$$

by $f \mapsto f(i)$.
We also have

$$k[x,y]/(x) \cong k(y)$$

by $f \mapsto f(0,y)$.
Lastly

$$k[x,y]/(x,y) \cong k$$

by $f \mapsto f(0,0)$.

Theorem 1.2. *Let* $I \subseteq A$ *be an ideal. There is a bijection*

$$\{\textit{ideals of } A/I\} \leftrightarrow \{\textit{ideals } J \subseteq A \textit{ such that } I \subseteq J\}.$$

The maps are

$$K \subseteq A/I \mapsto \pi^{-1}(K) \subseteq A,$$

and

$$J \subseteq A, I \subseteq J \mapsto J/I \subseteq A/I.$$

Proof. Let

$$\pi : A \to A/I$$

be the quotient map. First, if $K \subseteq A/I$ is an ideal, then $\pi^{-1}(K)$ is an ideal of A. Since $0 \in K$, and $\ker \pi = I$, we have

$$I \subseteq \pi^{-1}(K).$$

Conversely, suppose $J \subseteq A$ is an ideal with $I \subseteq J$. Then

$$J/I = \{j + I \mid j \in J\}$$

is an ideal of A/I. Now check that these constructions are inverse. If $K \subseteq A/I$, then

$$\pi(\pi^{-1}(K)) = K$$

because π is surjective. If $J \subseteq A$ contains I, then

$$\pi^{-1}(J/I) = J.$$

Indeed, $a \in \pi^{-1}(J/I)$ if and only if

$$a + I \in J/I,$$

which happens if and only if there exists $j \in J$ such that

$$a + I = j + I.$$

This means

$$a - j \in I.$$

Since $I \subseteq J$ and $j \in J$, we get $a \in J$. Hence

$$\pi^{-1}(J/I) = J.$$

Therefore the two maps are inverse bijections. $\square$

Example 1.6. *The ideals of $\mathbb{Z}/12\mathbb{Z}$ correspond to ideals of $\mathbb{Z}$ containing (12). The ideals of $\mathbb{Z}$ are precisely (n) for $n \geq 0$. We need $(12) \subseteq (n)$. This happens precisely when $n \mid 12$. So the ideals of $\mathbb{Z}/12\mathbb{Z}$ are*

$$(1), (2), (3), (4), (6), (12) = (0).$$

Proposition 1.5. *Let k be a field. The ideal*

$$(x, y) \subseteq k[x, y]$$

is not principal.

Proof. Suppose for a contradiction that

$$(x, y) = (f)$$

for some $f \in k[x, y]$. Since

$$x \in (x, y) = (f),$$

we have $f \mid x$. Since

$$y \in (x, y) = (f),$$

we have $f \mid y$. However, x and y have no common non-unit factor in $k[x, y]$. Hence f must be a unit. If f is a unit, then

$$(f) = (1) = k[x, y].$$

But this is impossible, because

$$1 \notin (x, y).$$

Therefore (x, y) is not principal. □

Thus $k[x]$ is a principal ideal domain when k is a field, but $k[x, y]$ is not.

Exercises

Problem 1.1. *Prove that if A is a ring with $1 = 0$, then A is the trivial ring.*

Proof. Let $a \in A$. Then

$$\begin{aligned} a &= a \\ 1 \cdot a &= 0 \cdot a \\ a &= 0. \end{aligned}$$

Thus

$$A = \{0\}.$$

□

Problem 1.2. *Let A be a ring. Prove that there is a unique homomorphism $\mathbb{Z} \to A$.*

Proof. Let A be a ring and let $\phi, \psi : \mathbb{Z} \to A$ be homomorphisms.

Then we require

$$\phi(1) = \psi(1) = 1_A.$$

Therefore, for $n > 0$, we have

$$\phi(n) = \phi\left(\sum_{k=1}^{n} 1\right) = \sum_{k=1}^{n} 1_A = \psi\left(\sum_{k=1}^{n} 1\right) = \psi(n).$$

For $n < 0$ we have $n = -m$ for some $m > 0$. Then

$$\phi(n) = \phi(-m) = -\phi(m) = -\sum_{k=1}^{n} 1_A = -\psi(m) = \psi(-m) = \psi(n).$$

Lastly,

$$\phi(0) = 0 = \psi(0).$$

Thus ϕ and ψ agree on all of $\mathbb{Z}$ and hence

$$\psi = \phi.$$

□

Problem 1.3. *Identify* $k[x, y]/(x, y - 1)$.

Solution. Let $\phi : k[x, y] \to k$ by

$$\phi(f) = f(0, 1).$$

Then ϕ is a ring homomorphism and furthermore it is surjective since, for $a \in k$ we have $f(x, y) := a \mapsto a$.

Now we show that $\ker\phi = (x, y - 1)$. First of all $x, y - 1 \in \ker\phi$ since $x(0, 1) = 0$ and $y - 1(0, 1) = 0$. Thus

$$(x, y - 1) \subseteq \ker\phi.$$

Now suppose $f \in \ker\phi$. Then $f(0, 1) = 0$ so $f(x, y) = xg(x, y) + (y - 1)h(x, y) + f(0, 1)$. Since $f(0, 1) = 0$ we have $f(x, y) = xg(x, y) + (y - 1)h(x, y)$ so $f(x, y) \in (x, y - 1)$. Thus $\ker\phi \subseteq (x, y - 1)$. The two together give $(x, y - 1) = \ker\phi$. Thus, by the fundamental theorem of ring homomorphisms (or first isomorphism theorem, if you prefer), we have

$$\frac{k[x, y]}{(x, y - 1)} \cong k.$$

□

Problem 1.4. *List all ideals of* $\mathbb{Z}/18\mathbb{Z}$.

Solution. The ideals are

- (0),
- (1),
- (2),
- (3),
- (6),
- (9),

i.e. (n) where $n \mid 18$. Note that $(0) = (18)$ and 18 divides 18. □

2 Special ideals and rings

Definition 2.1. *Let A be a ring.*
An element $u \in A$ is a ***unit*** *if there exists $v \in A$ such that*

$$uv = 1.$$

An element $x \in A$ is a ***zero-divisor*** *if there exists a nonzero element $y \in A$ such that*

$$xy = 0.$$

A ring A is an ***integral domain*** *if*

$$A \neq 0$$

and 0 is the only zero-divisor.
Equivalently, A is an integral domain if

$$ab = 0 \Rightarrow a = 0 \text{ or } b = 0.$$

A ring A is a ***field*** *if*

$$A \neq 0$$

and every nonzero element of A is a unit.

Example 2.1. *$\mathbb{Z}$ is an integral domain, but not a field. The rings*

$$\mathbb{Q},\ \mathbb{R},\ \mathbb{C}$$

are fields. The ring $\mathbb{Z}/4\mathbb{Z}$ is not an integral domain.

Proposition 2.1. *Let A be a ring and $x \in A$. Then*

$$x \text{ is a unit}$$

if and only if

$$(x) = (1).$$

Proof. Suppose first x is a unit. Then $\exists y \in A$ s.t.

$$xy = 1.$$

Therefore $1 \in (x)$. Since every ideal containing 1 is the whole ring, we get

$$(x) = A = (1).$$

Conversely, suppose $(x) = (1)$. Then

$$1 \in (x).$$

By definition of the principal ideal (x), there exists $y \in A$ such that

$$1 = yx.$$

Since A is commutative,

$$xy = 1.$$

Therefore x is a unit. $\square$

Theorem 2.1. *Let A be a ring. Then A is a field if and only if A has exactly two ideals:*

$$(0) \neq (1).$$

Proof. Suppose A is a field.

Let $I \subseteq A$ be a nonzero ideal. Choose

$$0 \neq x \in I.$$

Since A is a field, x is a unit. Then

$$(x) = (1).$$

But

$$(x) \subseteq I.$$

Therefore

$$(1) = A \subseteq I$$

so

$$I = (1).$$

So the only ideals are

$$(0), \ (1).$$

Conversely, suppose A has two ideals

$$(0) \neq (1).$$

Let $0 \neq x \in A$. Then

$$(x) \neq (0).$$

Since the only nonzero ideal is (1), we have

$$(x) = (1).$$

Then x is a unit and thus since every nonzero element of A is a unit, A is a field. $\square$

Definition 2.2 (Prime ideal). *Let A be a ring and let $I \subseteq A$ be an ideal. We say I is a* ***prime ideal*** *if*

$$I \neq (1)$$

and whenever

$$ab \in I,$$

we have

$$a \in I \quad \text{or} \quad b \in I.$$

Equivalently,

$$a, b \notin I \implies ab \notin I.$$

The set of all prime ideals of A is denoted

$$\text{Spec}(A).$$

Example 2.2. *The prime ideals of $\mathbb{Z}$ are (0) and (p) where p is a prime number.*

Definition 2.3 (Maximal ideal)**.** *Let A be a ring and let $I \subseteq A$ be an ideal. We say I is a* ***maximal ideal*** *if*

$$I \neq (1)$$

and the only ideals J satisfying

$$I \subseteq J \subseteq A$$

are

$$J = I$$

and

$$J = A.$$

Equivalently, there is no ideal strictly between I and A. The set of maximal ideals of A is denoted

$$\text{Specm}(A).$$

Example 2.3. *The maximal ideals of $\mathbb{Z}$ are (p).*

Theorem 2.2. *Let A be a ring and let $I \subseteq A$ be an ideal. Then*

$$I \text{ is prime}$$

if and only if

$$A/I \text{ is an integral domain.}$$

Also,

$$I \text{ is maximal}$$

if and only if
$$A/I \text{ is a field.}$$

Exercises

Problem 2.1. *Decide whether $(2) \subseteq \mathbb{Z}/6\mathbb{Z}$ is prime, maximal, both, or neither.*

Solution. It is maximal. Let $\phi : \mathbb{Z}/6\mathbb{Z} \to \mathbb{Z}/2\mathbb{Z}$ by
$$\phi(a) = a \mod 2.$$
Then this map is surjective and $\ker\phi = (2)$. By the first isomorphism theorem we have
$$\mathbb{Z}_6/\ker\phi \cong \mathbb{Z}_2$$
which is a field and thus $\ker\phi$ is maximal. □

Problem 2.2. *Decide whether $(x-1) \subseteq k[x]$ is prime or maximal.*

Solution. Let $\phi : k[x] \to k$ by
$$\phi(f) = f(1).$$
Then $\ker\phi = (x-1)$, the map is surjective so it follows as from before that $(x-1)$ is maximal. □

3 Krull's theorem, PIDs/UFDs, local rings

Theorem 3.1 (Krull's theorem). *Let A be a nonzero ring. Then A has a maximal ideal.*
Equivalently:
$$A \neq 0 \Rightarrow \mathrm{Specm}(A) \neq \emptyset.$$

This is not at all obvious for arbitrary rings, because rings may have infinitely many ideals and no obvious "largest" proper ideal. To prove Krull's theorem we will need the following set-theoretic principle.

Lemma 3.1 (Zorn's lemma). *Let S be a partially ordered set. Suppose every chain in S has an upper bound in S. Then S has a maximal element.*

Here:

- a **chain** is a subset where any two elements are comparable,
- an **upper bound** for a chian $C \subseteq S$ is an element $u \in S$ such that $c \leq u$ for all $u \in C$,
- a **maximal element** is an element $m \in S$ such that there is no $s \in S$ with $m < s$.

Proof of Krull's theorem. Let $A \neq 0$. Consider the set

$$S = \{I \subseteq A \mid I \text{ is an ideal and } I \neq A\}.$$

So S is the set of all proper ideals of A. We partially order S by inclusion:

$$I \leq J \Leftrightarrow I \subseteq J.$$

We want to apply Zorn's lemma.

Let $\{I_\lambda\}_{\lambda \in \Lambda}$ be a chain in S. Define

$$I = \bigcup_{\lambda \in \Lambda} I_\lambda.$$

We claim $I \in S$, meaning I is a proper ideal.

First, I is an ideal.

Let $a, b \in I$. Then there exists $\lambda, \mu \in \Lambda$ such that

$$a \in I_\lambda, \ \ b \in I_\mu.$$

Since I_μ is an ideal,

$$a - b \in I_\mu \subseteq I.$$

Thus I is an additive subgroup. Now let $r \in A$ and $a \in I$. Then $a \in I_\lambda$ for some λ, so

$$ra \in I_\lambda \subseteq I.$$

Thus I absorbs multiplication from A and is therefore an ideal. Second $I \neq A$. If $I = A$, then in particular

$$1 \in I.$$

So $1 \in I_\lambda$ for some λ, but then

$$I_\lambda = A,$$

contradicting $I_\lambda \in S$. Hence $I \neq A$.

So $I \in S$, and I is an upper bound for the chain. Therefore every chain in S has an upper bound. By Zorn's lemma, S has a maximal element m. This m is a proper ideal of A, and no proper ideal strictly contains it. Hence m is a maximal ideal.

Thus every nonzero ring has a maximal ideal. $\square$

Corollary 3.1. *Let $I \subseteq A$ be a proper ideal. Then there exists a maximal ideal $m \subseteq A$ such that*

$$I \subseteq m.$$

Proof. Since $I \neq A$, the quotient

$$A/I$$

is nonzero. By Krull's theorem, A/I has a maximal ideal $\mathfrak{m}$. Let

$$\pi : A \to A/I$$

be the quotient map. Then

$$m = \pi^{-1}(\mathfrak{m})$$

is a maximal ideal of A containing I, by the correspondence theorem. □

Corollary 3.2. *Let A be a ring and $x \in A$. Then*

$$x \text{ is a unit}$$

if and only if

$$x \notin m$$

for every maximal ideal $m \subseteq A$.
Equivalently:

$$x \text{ is not a unit} \Leftrightarrow x \in m \text{ for some maximal ideal } m.$$

Proof. We have:

$$x \text{ is not a unit} \Leftrightarrow (x) \neq A.$$

By the previous Corollary,

$$(x) \neq A \Leftrightarrow (x) \subseteq m$$

for some maximal ideal m. But

$$(x) \subseteq m \Leftrightarrow x \in m.$$

Therefore

$$x \text{ is not a unit} \Leftrightarrow x \in m$$

for some maximal ideal m. □

Definition 3.1 (PID). *An integral domain A is a* ***principal ideal domain****, or PID, if every ideal of A is principal.*

Example 3.1. *Some examples of PIDs include*

- $k[x]$ *for any field* k,
- *any field* k,
- $\mathbb{Z}$.

Definition 3.2. *Let A be an integral domain. Let $f \in A$ be neither 0 nor a unit. We say f is* ***irreducible*** *if whenever*

$$f = gh,$$

then either g is a unit or h is a unit, meaning f cannot be factored nontrivially.

Definition 3.3 (UFD). *An integral domain A is a* ***unique factorization domain***, *or UFD, if every nonzero nonunit $f \in A$ can be written as a finite product of irreducibles,*

$$f = p_1 \cdots p_n,$$

and this factorization is unique up to reordering and multiplication by units.

Example 3.2. *Some examples include:*

- $\mathbb{Z}$,
- $k[x]$ *for any field* k,
- $k[x_1, \dots, x_n]$ *for any field* k.

We remark that PID $\Rightarrow$ UFD and that if A is a UFD then $A[x]$ is a UFD.

Definition 3.4 (Local ring). *A ring A is* ***local*** *if it has exactly one maximal ideal. That maximal ideal is usually denoted*

$$\mathfrak{m}.$$

So a local ring is a pair $(A, \mathfrak{m})$, where $\mathfrak{m}$ is the unique maximal ideal.

Example 3.3. *Every field k is local since the only maximal ideal is (0).*
$\mathbb{Z}$ is a nonexample as it has maximal ideals (p) for every prime p.

Proposition 3.1. *Let A be a local ring with unique maximal ideal $\mathfrak{m}$. Then the set of units in A is*

$$A \setminus \mathfrak{m}.$$

Equivalently:

$$x \in \mathfrak{m} \Leftrightarrow x \text{ is not a unit.}$$

Proof. By the corollary to Krull's theorem, an element $x \in A$ is a unit if and only if it is contained in no maximal ideal. But A has exactly one maximal ideal, namely $\mathfrak{m}$. Therefore

$$x \text{ is a unit} \Leftrightarrow x \notin \mathfrak{m}.$$

So the units are exactly

$$A \setminus \mathfrak{m}.$$

□

Proposition 3.2. *Let A be a ring. Suppose the set of nonunits*

$$\mathfrak{m} = \{x \in A \mid x \text{ is not unit}\}$$

is an ideal. Then A is local, and $\mathfrak{m}$ is its unique maximal ideal.

Proof. Let $I \subseteq A$ be a proper ideal. We claim

$$I \subseteq \mathfrak{m}.$$

Indeed, if $x \in I$ were a unit, then

$$1 \in I,$$

because

$$x^{-1}x = 1.$$

Then $I = A$, contradicting that I is proper. So every element of I is a nonunit, hence

$$I \subseteq \mathfrak{m}.$$

Thus every proper ideal of A is contained in $\mathfrak{m}$. Since $\mathfrak{m}$ is itself a proper ideal, it is the unique maximal ideal. Therefore A is local. □

Exercises

Problem 3.1.

(a) *Let A be a ring, and let $f = a_n x^n + \cdots + a_0 \in A[x]$. Prove that if f is a unit of $A[x]$, then a_0 is a unit of A.*

(b) *Show that if a is a unit of A, then the constant polynomial*

$a \in A[x]$ *is a unit of* $A[x]$.

(c) Show that if A *is an integral domain, then the units of* $A[x]$ *are exactly the constant polynomials* a *with* a *a unit of* A.

Proof of (a). Suppose
$$f = \sum_{k=0}^{n} a_k x^k$$
is a unit of $A[x]$. Then there exists
$$g = \sum_{k=0}^{m} b_k x^k$$
such that
$$fg = 1.$$
Let $\phi : A[x] \to A$ by
$$\phi(f) = f(0).$$
Then ϕ is a ring homomorphism, and thus
$$1_A = \phi(1_{A[x]}) = \phi(fg) = a_0 b_0$$
so a_0 is a unit in A. □

Proof of (b). Suppose $a \in A$ is a unit. Then there exists $b \in A$ so that
$$ab = 1_A.$$
Then $a, b \in A[x]$ as constant polynomials. Once again
$$1_{A[x]} = 1_A = ab = \phi(a)\phi(b) = \phi(ab).$$
□

Proof. From the above we have that units in A as constant polynomials are units in $A[x]$.

Now let $f \in A[x]$ be a unit. Then there is $g \in A[x]$ so
$$fg = 1.$$
Then since $\deg(g \cdot h) = \deg(g) + \deg(h)$ we have
$$\deg(f) + \deg(g) = \deg(f \cdot g) = \deg(1) = 0.$$
Then $\deg(f) = \deg(g) = 0$ so they are constant polynomials.

We require A to be an integral domain because otherwise we could take $A = \mathbb{Z}_4$ and $f = 2x + 1$. Then $f^2 = 1$. □

Problem 3.2. *Prove that a ring A is a field if and only if the following holds: $A \neq 0$ and every homomorphism from A to a non-zero ring is injective.*

Proof. Let A and $B \neq 0$ be rings.

First assume A is a field. Then $\ker \phi$ is A or (0) for every homomorphism $\phi : A \to B$. If $\ker \phi = A$ then $\psi(x) = 0_B$ for all $x \in A$ and in particular $1_B = \phi(1_A) = 0_B$ so B is the trivial ring, but this contradicts our assumption. Thus $\ker \phi = (0)$, meaning ϕ is injective. Furtheremore, by definition, $A \neq 0$.

Now assume $A \neq 0$ and every homomorphism $\phi : A \to B$ has $\ker \phi = \{0_A\}$ when B is nonzero.

Let $I \subseteq A$ is some ideal. Consider the canonical quotient homomorphism

$$\pi : A \to A/I \text{ defined by } \pi(a) = a + I.$$

Now either A/I is nonzero or not. If it is the zero ring then $\ker \pi = A$. Otherwise if A/I is nonzero, then by assumption $\ker \pi = (0)$. By construction,

$$\ker \pi = I$$

so every ideal I of A is either (0) or A. Hence A is a field. □

Problem 3.3. *Let A be a ring, and let $x \in A$. Consider the map $m_x : A \to A$ given by $m_x(y) = xy$ for some $x \in A$.*

(a) Show that m_x is a ring homomorphism if and only if $x = 1$.

(b) Show that x is a 0-divisor if and only if m_x is non-injective.

(c) Show that x is a unit if and only if m_x is surjective.

Proof of (a). Suppose $x = 1$. Then $m_x : A \to A$ is the identity map $\mathrm{id} : A \to A$,

$$\mathrm{id}(y) = y,$$

which is a homomorphism.

Now suppose m_x is a ring homomorphism and suppose for a contradiction that $x \neq 1$. Then

$$m_x(1) = x \neq 1$$

so m_x does not preserve the multiplicative identity and thus fails to be a homomorphism. □

Proof of (b). Suppose x is a 0-divisor, meaning $x \neq 0$ and there exists $\xi \in A \setminus \{0\}$ so that

$$x \cdot \xi = 0.$$

Then $\xi \in \ker m_x$ and $\xi \neq 0$ so m_x cannot be injective.

Suppose m_x is not injective. So $|\ker m_x| > 1$. Then there exists $\xi \in \ker m_x \setminus \{0\}$. Then

$$\begin{aligned} m_x(\xi) &= m_x(0) \\ x \cdot \xi &= x \cdot 0 = 0 \end{aligned}$$

so x is a 0-divisor. □

Proof of (c). Suppose x is a unit. Then there exists $x^{-1} \in A$ so that $xx^{-1} = 1$. Then, for any $a \in A$ we see that

$$xx^{-1}a \overset{m_x}{\mapsto} a$$

so m_x is surjective.

Now suppose m_x is surjective. Then, for any $a \in A$ there exists $b \in A$ so that

$$m_x(b) = xb = a.$$

In particular this holds for $1 \in A$ so there exists $x^{-1} \in A$ so that

$$m_x(x^{-1}) = xx^{-1} = 1.$$

□

Problem 3.4. *Let A be a ring, and let $\{P_i\}_{i \in I}$ be a non-empty chain of prime ideals (where "chain" ,eans that for every two $i, j \in I$ either $P_i \subseteq P_j$ or $P_j \subseteq P_i$). Prove that $\bigcup_{i \in I} P_i$ and $\bigcap_{i \in I} P_i$ are prime ideals.*

Proof. Let

$$J = \bigcup_{i \in I} P_i$$

and

$$K = \bigcap_{i \in I} P_i.$$

We begin with J. Since the chain is nonempty, pick any P_i. Since P_i is an ideal, $0 \in P_i$ so $0 \in J \neq \emptyset$. Let $x \in J$ and $r \in A$. Since $x \in J$, $x \in P_i$ for some specific index i. Since P_i is an ideal we have $rx \in P_i$ so $rx \in J$. Let $x, y \in J$. Then $x \in P_i$ and $y \in P_j$ for some specific indicies $i, j \in I$. The chain-condition gives us that $P_i \subseteq P_j$ or vice-versa. Assume wlog the former holds. Then $x, y \in P_j$ so $x + y \in P_j \subseteq J$. Thus J is closed under addition.

Now since every P_i is prime we have $1 \notin P_i$ for every $i \in I$ so $1 \notin J$. Thus $J \neq A$ and is therefore a proper ideal. Let $a, b \in A$ such that $ab \in J$. Then there is some P_i so that $ab \in P_i$. Since P_i is prime we have $a \in P_i$ or $b \in P_i$.

Now we do K. Similar as before $0 \in P_i$ for all $i \in I$ so $0 \in K \neq \emptyset$. Let $x \in J$ and $r \in A$. Since $x \in J$ we have that $x \in P_i$ for every i so $ax \in P_i$ for every i hence $ax \in K$. Let $x, y \in K$. Then $x, y \in P_i$ for every i hence $x + y \in P_i$ for every i so $x + y \in K$. Thus K is closed under addition.

Now supose $ab \in K$. Then $ab \in P_i$ for every i so $a \in P_i$ or $b \in P_i$ for every i. Thus $a \in K$ or $b \in K$. □

Problem 3.5. *Let X be a topological space and $x \in X$. We consider pairs (U, f), where $U \subseteq X$ is an open set with $x \in U$ and $f : U \to \mathbb{R}$ is a continuous function.*
Two pairs (U, f) and (V, g) are called equivalent if there exists an open neighborhood W of x with $W \subseteq U \cap V$ such that $f\mid_W = g\mid_W$. An equivalence class is called a germ of a function at x. We denote by $\mathcal{O}_{X,x}$ the set of all such germs.

(a) *Show that addition and multiplication of germs are well defined by*
$$(U, f) + (V, g) := (U \cap V, f\mid_{U\cap V} + g\mid_{U\cap V}),$$
$$(U, f) + (V, g) := (U \cap V, f\mid_{U\cap V}\, g\mid_{U\cap V}).$$
Verify that $\mathcal{O}_{X,x}$ is a ring with these operations, and describe the elements $0, 1 \in \mathcal{O}_{X,x}$ as germs.

(b) *Show that a germ (U, f) is a unit in $\mathcal{O}(X, x)$ if and only if $f(x) \neq 0$.*

(c) *Show that $\mathcal{O}_{X,x}$ is a local ring, with maximal ideal*
$$\mathfrak{m} = \{(U, f) \in \mathcal{O}_{X,x} \mid f(x) = 0\},$$
and residue field
$$\mathcal{O}_{X,x}/\mathfrak{m} \cong \mathbb{R}.$$

Proof of (a). Suppose $(U_1, f_1) = (U_2, f_2)$ and $(V_1, g_1) = (V_2, g_2)$. By definition of the equivalence relation

- there exists an open neighborhood $W_1 \subseteq U_1 \cap U_2$ of x such that $f_1\mid_{W_1} = f_2\mid_{W_1}$, and
- there exists an open neighborhood $W_2 \subseteq V_1 \cap V_2$ of x such that $g_1\mid_{W_2} = g_2\mid_{W_2}$.

Define $W = W_1 \cap W_2$. Since open neighborhoods are closed under finite intersections we have that W is an open neighborhood of x. Furthermore, $W \subseteq (U_1 \cap v_1) \cap (U_2 \cap V_2)$. Restricting the sum to W we

get
$$(f_1+g_2)\big|_W = f_1\big|_W + g_1\big|_W + f_2\big|_W + g_2\big|_W = (f_2+g_2)\big|_W .$$
Thus addition is well-defined and this approach works for multiplication as well.

The ring properties of associativity, commutativity and distributivity are inherited from the pointwise operations of $\mathbb{R}$.

The additive itentity is $(X, \mathbf{0})$ where $\mathbf{0} : X \to \mathbb{R}$ is the constant function 0.

The multiplicative identity is $(X, \mathbf{1})$ where $\mathbf{1} : X \to \mathbb{R}$ is the constant 1. $\square$

Proof of (b). Suppose (U, f) is a unit. Then there exists (V, g) such that $(U, f) \cdot (V, g) = (X, \mathbf{1})$. Then there is an open neighborhood $W \subseteq U \cap V$ of x where $f(k)g(k) = 1$ for all $k \in W$. Evaluated at x we get
$$f(x)g(x) = 1.$$
Since $f(x), g(x) \in \mathcal{R}$ we cannot have $f(x) = 0$ as 0^{-1} does not exist.

Suppose $f(x) \neq 0$. Since $f : U \to \mathbb{R}$ is continuous and $\{0\}^c = \mathbb{R} \setminus \{0\}$ is an open subset of $\mathbb{R}$, the preimage $V = f^{-1}(\{0\}^c)$ is open in U and thus open in X.

Since $f(x) \neq 0$ we have $x \in V$ so we can define a continuous $g : V \to \mathbb{R}$ by
$$g(k) = \frac{1}{f(k)} \text{ for all } k \in V.$$
Then, on the open neighborhood V, we have $f\big|_V \cdot g = 1\big|_V$. Thus $[U, f]$ is a unit. $\square$

4 A step back

Everything thus far has been loosely covering Jørgen V. Rennemo's notes as I was planing to take this course at UiO as I usually do. Now I will be taking the equivalent MAT224 course at UiB instead! Thus I think it is fitting to go back, as there are some pretty big differences in the flavour of this course when compared with MAT4200 of UiO.

Gathmann motivates commutative algebra by telling us that commutative rings provide a common algebraic language for things that initially look number-theoretic. In particular, polynomial equations define geometric objects, and the functions on those objects form commutative rings.

The philosophy is that instead of studying geometric objects directly, study the ring of polynomial functions on it. This is quite a bit more algebraic geometry esque. Very cool!

Definition 4.1. *Let K be a field. The affine n-space over K is the set*

$$\mathbb{A}^n_K = \{(a_1, \dots, a_n) \mid a_i \in K\}.$$

As a set,

$$\mathbb{A}^n_K = K^n.$$

The different notation emphasizes that we are not interested in the set as a vector space. We are thinking of it simply as a set of points on which polynomial equations can be imposed.

For instance $\mathbb{A}^2_{\mathbb{R}}$ is simply the real plane.

Let $f \in K[x_1, \dots, x_n]$. Write

$$f = \sum_{(i_1,\dots,i_n)\in\mathbb{N}^n} a_{i_1\dots,i_n} x_1^{i_1} \cdots x_n^{i_n},$$

where only finitely many coefficients are nonzero.

Given $a = (a_1, \dots, a_n) \in \mathbb{A}^n_K$, we evaluate f at a by

$$f(a) = \sum a_{i_1,\dots,i_n} a_1^{i_1} \cdots a_n^{i_n}.$$

The distinction is simply between the formal variables x_j and a point with coordinates a_j at which we substitute values.

Definition 4.2. *Let*

$$S \subseteq K[x_1, \dots, x_n]$$

be any set of polynomials. Its zero locus is

$$V(S) = \{a \in \mathbb{A}^n_K \mid f(a) = 0 \textit{ for every } f \in S\}.$$

A subset of $\mathbb{A}^n_K$ that can be written as $V(S)$ is called an ***affine***

variety *using Gathmann's terminology.*

If $S = (f_1, \dots, f_r)$ we normally write $V(f_1, \dots, f_r)$.

Example 4.1. *Over* $\mathbb{R}$,

$$V(x^2 + y^2 - 1) \subseteq \mathbb{A}^2_{\mathbb{R}}$$

is the unit circle. Indeed,

$$(a, b) \in V(x^2 + y^2 - 1)$$

if and only if

$$a^2 + b^2 = 1.$$

Example 4.2. *Consider*

$$V(xy) \subseteq \mathbb{A}^2_{\mathbb{K}}.$$

A point (a, b) *lies in* $V(xy)$ *iff* $ab = 0$. *Since* K *is a field, this means* $a = 0$ *or* $b = 0$. *Hence*

$$V(xy) = V(x) \cup V(y).$$

Geometrically this is the union of the x *and* y *axes. This notion will appear later algebraically through prime ideals and primary decomposition. How exciting!*

Now, why can't we simply go through equations themselves? Suppose $X = V(f)$. It would be tempting to say that f is the equations of X. However, there is an immediate problem:

$$V(f) = V(f^2).$$

Many entirely different collectioons of polynomials can have exactly the same zero locus. Gathmann uses this ambiguity to motivate replacing a chosen set of equations by all equations that vanish on the object.

Definition 4.3. *Let* $X \subseteq \mathbb{A}^n_K$. *We do not actuaklly need to assume* X *is a variety yet.*
The ideal of X *is*

$$I(X) = \{f \in K[x_1, \dots, x_n] : f(a) = 0 \text{ for every } a \in X\}.$$

Proposition 4.1.

$$I(x) \lhd K[x_1, \dots, x_n].$$

Proof. Ceirtainly $0 \in I(X)$.

Let $f, g \in I(X)$. For every $a \in X$,

$$(f-g)(a) = f(a) - g(a) = 0 - 0 = 0.$$

Hence $f - g \in I(X)$. Thus $I(X)$ is an additive subgroup. Now let $h \in K[x_1, \dots, x_n]$ and $f \in I(X)$. For every $a \in X$,

$$(hf)(a) = h(a)f(a) = h(a) \cdot 0 = 0.$$

Thus $hf \in I(X)$. Therefore $I(X)$ is an ideal. □

We now take the quotient

$$A(X) = K[x_1, \dots, x_n]/I(X).$$

This is called the **coordinate ring** of X, or the ring of polynomials functions on X.

Why quotient by $I(X)$?

Suppose $f, g \in K[x_1, \dots, x_n]$. In the quotient, $f + I(X) = g + I(X)$ if and only if $f - g \in I(X)$ so $f(a) = g(a)$ for every $a \in X$.

Thus $f = g$ in $A(X)$ if and only if f and g define the same function on X.

Example 4.3. *Let $X = V(y - x^2) \subseteq \mathbb{A}^n_K$.*
On X we have $y = x^2$. Thus as polynomial functions on X, y and x^2 are the same function.
Algebraically this is encoded by $A(X) = K[x, y]/I(X)$.

There is some nuance here when working over finite fields, but Gathmann says the course will largely surpress this as we will be working over infinite fields like $\mathbb{R}$ and $\mathbb{C}$ for the most part.

We will now introduce the ideal of a point.

Take $a = (a_1, \dots, a_n)$ in the affine n-space over some field K. Then

$$I(a) = I(\{a\}) = (x_1 - a_1, \dots, x_n - a_n).$$

Proposition 4.2. *For a as above, we have*

$$I(a) = (x_1 - a_1, \dots, x_n - a_n).$$

Proof. First suppose $f \in (x_1 - a_1, \dots, x_n - a_n)$.

Then there exist polynomials $g_1, \dots, g_n$ such that

$$f = \sum_{i=1}^{n} (x_i - a_i) g_i.$$

Evaluating at a,

$$f(a) = \sum_{i=1}^{n} (a_i - a_i) g_i(a) = 0.$$

Hence $f \in I(a)$. Therefore

$$(x_1 - a_1, \dots, x_n - a_n) \subseteq I(a).$$

For the reverse, let $f \in I(a)$, so

$$f(a_1, \dots, a_n) = 0.$$

We can write

$$\begin{aligned} f(x_1, \dots, x_n) - f(a_1, \dots, a_n) &= (f(x_1, x_2, \dots, x_n) - f(a_1, x_2, \dots, x_n)) \\ &+ (f(a_1, x_2, \dots, x_n) - f(a_1, a_2, x_3, \dots, x_n)) \\ &+ \cdots \qquad + (f(a_1, \dots, a_{n-1}, \end{aligned}$$

The first summand is divisible by $x_1 - a_1$, the second by $x_2 - a_2$, and so on. Therefore

$$f - f(a) \in (x_1 - a_1, \dots, x_n - a_n).$$

But $f(a) = 0$, so

$$f \in (x_1 - a_1, \dots, x_n - a_n)$$

giving us the inclusion. Together

$$I(a) = (x_1 - a_1, \dots, x_n - a_n).$$

□

Immediately, the evaluation map

$$\mathrm{ev}_a : K[x_1, \dots, x_n] \to K, \ \ f \mapsto f(a)$$

has

$$\ker \mathrm{ev}_a = I(a).$$

It is surjective so

$$\frac{K[x_1, \dots, x_n]}{I(a)} \cong K.$$

So far our ideals lived in $K[x_1, \dots, x_n]$. Suppose X is already a variety.

If $Y \subseteq X$, we can define

$$I_X(Y) = \{ f \in A(X) \mid f(y) = 0 \text{ for every } y \in Y \}.$$

Conversely, if $S \subseteq A(X)$, define

$$V_X(S) = \{ x \in X \mid f(x) = 0 \text{ for every } f \in S \}.$$

We surpress subscripts when the variety is clear.

Proposition 4.3. *If $Y \subseteq Y' \subseteq X$, then*
$$I(Y') \subseteq I(Y).$$

The proof of this is quite short so I won't bother writing it. +

Likewise, if
$$S \subseteq S' \subseteq A(X),$$
then
$$V(S') \subseteq V(S).$$

Intuitively, larger geometry has a direct correspondence with a smaller ideal, and at the extremes
$$\mathbb{A}^n_K = (0)$$
in the usual infinite-field situation, while
$$I(\emptyset) = K[x_1, \dots, x_n] = (1),$$
since every polynomials vacuously vanishes on the empty set.

Suppose $Y \subseteq X$.

We can form
$$Y \mapsto I(Y) \mapsto V(I(Y)).$$

Ceirtainly,
$$Y \subseteq V(I(Y)).$$
Every polynomials in $I(Y)$ was defined precisely to vanish on Y. Similarly, for $S \subseteq A(X)$,
$$S \subseteq I(V(S)).$$

If Y is already a subvariety, something stronger holds.

Proposition 4.4. *If $Y \subseteq X$ is a subvariety, then*
$$V(I(Y)) = Y.$$

Proof. We know $Y \subseteq V(I(Y))$. Because Y is a subvariety, there exists some $S \subseteq A(X)$ such that
$$Y = V(S).$$
Every element of S vanishes on Y, so
$$S \subseteq I(Y).$$

Since taking zero loci reverses inclusion,
$$V(I(Y)) \subseteq V(S) = Y.$$

□

Now for a really nice result.

Let $Y \subseteq X$ be a subvariety.

Any polynomial function on X can be restricted to Y. Thus we get a ring homomorphism

$$\rho A(X) \to A(Y), \;\; f \mapsto f\mid_Y .$$

Proposition 4.5. *The map ρ is surjective and*

$$\ker \rho = I(Y).$$

Hence

$$A(X)/I(X) \cong A(Y).$$

Proof. By the definition of polynomial functions on Y, every polynomial function on Y is obtained by restricting a polynomial function from the ambient space, so ρ is surjective. Furthermore,

$$f \in \ker \rho$$

if and only if

$$f\mid_Y = 0,$$

which means precisely that

$$f(y) = 0$$

for every $y \in Y$.

Thus $\ker \rho = I(Y)$. The first isomorphism theorem then gives

$$A(X)/I(Y) = \frac{K[x_1, \dots, x_n]/I(X)}{I(Y)} \cong A(Y).$$

□

The last major thing now concerns maps between varieties.

Let $X \subseteq \mathbb{A}^n_K$ and $Y \subseteq \mathbb{A}^m_K$ be varieties.

Definition 4.4. A ***morphism of varieties***

$$f : X \to Y$$

is a map given by polynomials:

$$f(x) = (f_1(x), \dots, f_m(x)),$$

where

$$f_i \in K[x_1, \dots, x_n].$$

Gathmann then associates to f a ring homomorphism

$$f^* : A(Y) \to A(X)$$

defined by

$$f^*(g) = g \circ f.$$

Notice the direction:

$$X \xrightarrow{f} Y$$

produces

$$A(Y) \xrightarrow{f^*} A(X).$$

The cool thing now is that the arrow reverses.

Suppose you have an ordinary function $f : X \to Y$ and a real-valued function $g : Y \to \mathbb{R}$. Composition gives $g \circ f : X \to \mathbb{R}$.

So a map of spaces $X \to Y$ naturally lets you pull functions backwards

$$\{\text{functions on } Y\} \to \{\text{functions on } X\}.$$

This is called **pullback**.

Hence the notation f^*. This contravariance is very important further into algebraic geometry.

4.1 A few practice problems

Problem 4.1. *Determine geometrically*

$$V(xy, xz) \subseteq \mathbb{A}^3_K.$$

Try to describe it as a union of familiar linear pieces.

Solution. We have

$$V(xy, xz) = \{(x, y, z) \in K^3 : xy = 0, xz = 0\}.$$

Both equations share a common factor of x. If $x = 0$, then both equations vanish, so this gives the plane

$$V(x).$$

If $x \neq 0$, then since K is a field we may cancel x to get $y = 0$, $z = 0$, giving us the x-axis $V(y, z)$. Therefore

$$V(xy, xz) = V(x) \cup V(y, z).$$

Geometrically this is the union of the yz-plane and x-axis. $\square$

Problem 4.2. *Let* $a = (2, -1) \in \mathbb{A}^2_{\mathbb{R}}$. *Without a long proof, determine*

$I(a)$ and identify

$$\mathbb{R}[x,y]/I(a)$$

Solution. We showed earlier that

$$I(a) = (x-2, y+1).$$

Define the ring homomorphism

$$\phi : \mathbb{R}[x,y] \to \mathbb{R}, \quad f \mapsto f(2,-1).$$

The map is surjective and $\ker\phi = I(a)$. Thus

$$\mathbb{R}[x,y]/I(a) \cong \mathbb{R}.$$

$\square$